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Set 3: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and

labels.

2. Generate a bar chart showing the distribution of employees across different departments. Label the

chart elements.

3. Build a scatter plot to analyze the correlation between years of service and performance scores.

Explain any insights.

4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

SOLUTION :

CODE :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as gg
from plotly.subplots import make_subplots

df_employee = pd.DataFrame({
    'Employee_ID': [1, 2, 3, 4, 5],
```

```
'Department': ['Sales', 'HR', 'Marketing', 'Sales', 'HR'],  
'Years_of_Service': [5, 3, 7, 4, 2],  
'Performance_Score': [85, 92, 78, 90, 76]  
})
```

```
monthly_perf = pd.DataFrame({  
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May'],  
    'Sales_Dept': [80, 82, 85, 84, 88],  
    'HR_Dept': [88, 89, 90, 91, 92],  
    'Marketing_Dept': [75, 76, 75, 77, 78]  
})
```

```
plt.figure(figsize=(8, 4))  
plt.plot(monthly_perf['Month'], monthly_perf['Sales_Dept'], marker='o', label='Sales')  
plt.plot(monthly_perf['Month'], monthly_perf['HR_Dept'], marker='s', label='HR')  
plt.plot(monthly_perf['Month'], monthly_perf['Marketing_Dept'], marker='^',  
label='Marketing')  
plt.title('Employee Performance Trend Over Time', fontsize=14)  
plt.xlabel('Month', fontsize=12)  
plt.ylabel('Average Performance Score', fontsize=12)  
plt.legend(title='Department')  
plt.grid(True, linestyle='--', alpha=0.6)  
plt.tight_layout()  
plt.show()
```

```
dept_counts = df_employee['Department'].value_counts()  
plt.figure(figsize=(8, 4))  
plt.bar(dept_counts.index, dept_counts.values, color=['#4c72b0', '#55a868', '#c44e52'],  
edgecolor='black')  
plt.title('Employee Distribution Across Departments', fontsize=14)
```

```
plt.xlabel('Department', fontsize=12)
plt.ylabel('Number of Employees', fontsize=12)
for i, v in enumerate(dept_counts.values):
    plt.text(i, v + 0.05, str(v), ha='center', fontweight='bold')
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(8, 4))
sns.scatterplot(data=df_employee, x='Years_of_Service', y='Performance_Score',
hue='Department', s=150)
plt.title('Years of Service vs. Performance Score', fontsize=14)
plt.xlabel('Years of Service', fontsize=12)
plt.ylabel('Performance Score', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

```
print("--- Insights from Scatter Plot ---")
print("In this dataset, there is no direct positive correlation between years of service and performance scores.")
print("For instance, an employee with 3 years of service achieved a 92 performance score, while one with 7 years achieved 78.")
print("This indicates that tenure alone does not guarantee higher performance in this sample group.\n")
```

```
fig = make_subplots(
    rows=1, cols=2,
    subplot_titles=("Performance Trend Over Time", "Department Headcount")
)
```

```
fig.add_trace(gg.Scatter(x=monthly_perf['Month'], y=monthly_perf['Sales_Dept'],
name='Sales'), row=1, col=1)

fig.add_trace(gg.Scatter(x=monthly_perf['Month'], y=monthly_perf['HR_Dept'], name='HR'),
row=1, col=1)

fig.add_trace(gg.Scatter(x=monthly_perf['Month'], y=monthly_perf['Marketing_Dept'],
name='Marketing'), row=1, col=1)

fig.add_trace(gg.Bar(x=dept_counts.index, y=dept_counts.values, marker_color='indigo',
name='Count'), row=1, col=2)

fig.update_layout(title_text="Set 3: Employee Performance Dashboard", height=450)

fig.show()
```

OUTPUT :



